Packet Tracer – Cabling Types Hands-On

Objectives

Part 1: Build a Small Network Topology

Part 2: Connect Devices Using the Correct Cable Types

Part 3: Verify Physical Connectivity

Part 4: Understand When to Use Straight-Through, Cross-Over, and Console Cables

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Background / Scenario

One of the first skills every network technician must master is selecting the correct cable for connecting network devices. Using the wrong cable may prevent devices from communicating properly.

In this lab, you will practice connecting devices using three common cable types found in Cisco Packet Tracer:

- Copper Straight-Through
- Copper Cross-Over
- Console Cable

You will also verify that the physical connections are working correctly and learn the purpose of each cable type.

This lab focuses on:

- Identifying common network cables
- Connecting different types of devices
- Understanding physical layer communication
- Accessing a network device through the console port

You are building an essential foundation for future switching and routing labs.

Instructions

Part 1: Build the Network Topology

Step 1: Add the Devices

a. Open **Cisco Packet Tracer**.

b. From **End Devices**, drag two **PCs** onto the workspace.

c. From **Network Devices → Switches**, drag one **2960 Switch** onto the workspace.

d. Rename the devices:

- PC0 → **PC-A**
- PC1 → **PC-B**
- Switch0 → **S1**

Part 2: Connect Devices Using Different Cable Types

Step 1: Connect PC-A to the Switch

A PC connecting to a switch requires a **Copper Straight-Through** cable.

a. Click **Connections** (lightning bolt icon).

b. Select **Copper Straight-Through**.

c. Connect:

- PC-A → FastEthernet0
- S1 → FastEthernet0/1

Wait until the link lights turn **green.**

Step 2: Connect PC-A Directly to PC-B

Two similar devices connected directly require a **Copper Cross-Over** cable.

a. Select **Copper Cross-Over**.

b. Connect:

- PC-A → FastEthernet0
- PC-B → FastEthernet0

If Packet Tracer does not allow the second connection because PC-A is already connected, temporarily remove the Straight-Through cable and create the direct PC-to-PC connection instead.

Wait for the link lights to turn **green**.

Step 3: Connect the Console Cable

A Console cable is used to configure network devices through the CLI.

a. Delete the Cross-Over cable if necessary.

b. Select the **Console** cable (light blue cable).

c. Connect:

- PC-A → RS-232
- S1 → Console

Unlike Ethernet cables, the Console cable will not display green link lights because it is used for device management rather than network communication.

Why Are Different Cable Types Needed?

Different devices transmit and receive data on different wire pairs.

- **Straight-Through Cable**
  - Connects unlike devices.
  - Examples:
    - PC ↔ Switch
    - Router ↔ Switch
- **Cross-Over Cable**
  - Connects like devices.
  - Examples:
    - PC ↔ PC
    - Switch ↔ Switch

        - Router ↔ Router
- **Console Cable**
    - Provides out-of-band management access.
    - Used to configure routers and switches through the CLI.
    - Does not carry normal network traffic.

Part 3: Verify the Console Connection

Step 1: Open the Terminal

a. Click **PC-A**.

b. Go to the **Desktop** tab.

c. Click **Terminal**.

A configuration window appears.

Leave the default settings and click **OK**.

You should see:

Switch>

This indicates that the console connection is working correctly.

Step 2: Access Privileged EXEC Mode

Type:

enable

The prompt should change to:

Switch#

You have successfully connected to the switch through the Console cable.

Part 4: Identify the Correct Cable

Use the following table as a quick reference:

Device 1	Device 2	Correct Cable
PC	Switch	Straight-Through
Router	Switch	Straight-Through
PC	PC	Cross-Over
Switch	Switch	Cross-Over
Router	Router	Cross-Over
PC (RS-232)	Switch Console	Console
PC (RS-232)	Router Console	Console

Troubleshooting Tips

If the connection does not work:

✓ Verify you selected the correct cable type.

✓ Make sure you connected the correct interfaces.

✓ Wait a few seconds for Ethernet interfaces to initialize.

✓ Ensure the link lights are green for Ethernet connections.

✓ Remember that Console cables do not produce green link lights.

✓ If the Terminal window is blank, close it and reconnect the Console cable.

What You Learned

- The difference between Straight-Through, Cross-Over, and Console cables.
- Which devices require each cable type.
- How to physically connect network devices in Packet Tracer.
- How to access a switch using the Console port.
- That Console connections are used for management, not normal network traffic.

Challenge Extensions (Optional)

Try modifying the lab:

1. Connect two switches together.
    - Which cable should you use?
2. Connect a router to the switch.
    - Which cable is required?
3. Try connecting two PCs with a Straight-Through cable.
    - Does the connection work? Why or why not?
4. Use the Console cable to connect to a router instead of a switch.
5. Build the following topology using the correct cables:

```
PC-A -------- S1 -------- Router1
 |
 |
Console
 |
 |
Management PC
```

Identify the cable type used for each connection before building it.